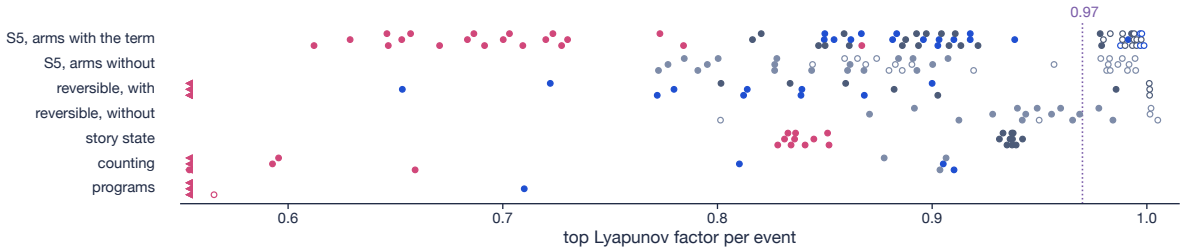
a **The loss ladder: one disc per seed, filled when exact at the test length**

		reversible programs 16 to 4,096 (256×)	S5 word problems 16 to 1,024 (64×)	programs 16 to 1,024 (64×)	Lean chains 8 to 1,024 (128×)	addition 32 to 4,096 (128×)	counting 32 to 4,096 (128×)	story state 16 to 1,024 (64×)
ordinary	O	0/3	2/3	0/3	0/3	0/3	0/3	0/3
+ fixed norm	O+N	2/3	2/10	0/3	3/3	3/3	1/3	0/3
+ centred closure	O+Nc	3/3	3/10	0/3	3/3	3/3	1/3	0/3
+ next-state, norm	NS+N	9/10	13/20	2/10	10/10	9/10	10/10	0/10
+ equivalence, norm	EQ+N	3/3	2/3	3/3	3/3	3/3	3/3	3/3
+ equivalence, centred	EQ+Nc		12/20					10/10
earlier recipe, no norm	FULL	0/3	2/3	0/3	3/3	0/3	0/3	0/3
earlier recipe, fixed norm	FULL+N	3/3	7/10	3/3	3/3	3/3	3/3	3/3
earlier recipe (NS+EQ)	FULL+Nc	10/10	15/20	8/10	10/10	10/10	10/10	10/10
ours (M): + aimed contraction	AIM+Nc	10/10	20/20	9/10	10/10	10/10	10/10	10/10

Filled: exact at the test length (≥ 0.99 of 512 problems at every test length). The method: 79 of 80 seeds exact where it has run; on Lean it uses the fixed-norm closure.

b **With the equivalence term, runs fail only where contraction is lost**



S5: all 16 inexact runs of arms with the equivalence term lie above 0.97 (so do 4 exact ones, near the edge); 11 of 22 inexact runs without the term lie below it: they fail by anchoring. The method's 20 S5 seeds lie at 0.61 to 0.87. Off the axis to the left: addition: all 16 runs below 0.30.
Filled: exact; open: not. Rose: the method (M); cobalt: our earlier recipe (NS+EQ); dark grey: the other arms with the equivalence term; light grey: arms without it.